

Optimizing Pathfinding Model Development for Indoor Robots

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INTRODUCTION

Autonomous systems have become more commonplace in recent years. It is thrilling to witness automation take on a greater role in our lives, such as with self-driving vehicles. That said, there is much to be done before fully integrating autonomous navigation into daily life. Hence, a huge focus of recent research in machine learning has been to fine tune and optimize systems to be robust, safe, and useful.

In order to assess the ability of these systems and continue to develop them, we aim to implement metrics that will give a clearer, more concrete rating on machine models and their performance. Ultimately, each model's performance depends on a combination of factors including:

- Differences in datasets
- Its architecture
- The training algorithm used for the model

By analyzing many trials and their results, we hope to determine the optimal conditions for producing a model that achieves the best outcomes.

DATA COLLECTION

Real-World Simulation

Using Blender, our team created a virtual environment of the hallways of Oldenburg (a dormitory at Pomona College). Then, we used the game-development platform Unreal Engine 5 to generate visuals as our simulated robot, or the "agent", progresses through the simulation. For some datasets, we periodically changed the textures of the floors, walls, and ceilings in real-time within the environment, which is important for collecting more diverse data.

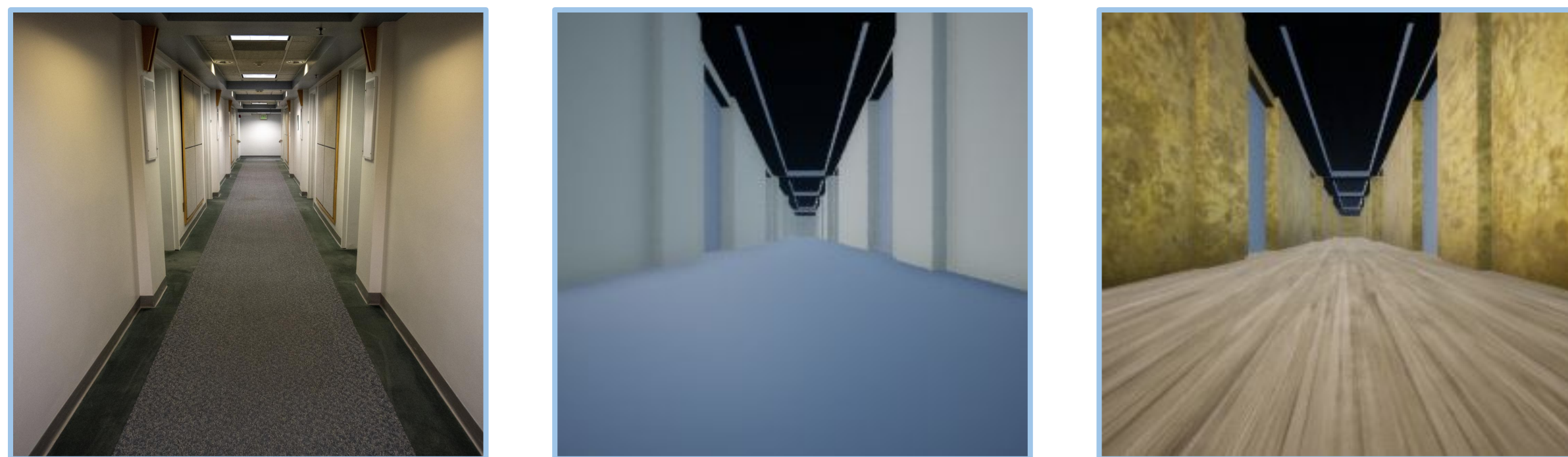


Figure 1: (left) Image of hallway in Oldenburg and (middle/right) simulated depictions of the environment

What is Box Navigation?

Box Navigation is a pathfinding system that uses overlapping boxes and targets to inform the agent of the environment's layout and where it should travel to next. It uses one of three unique navigation behaviors to determine its next action for the sake of varying data. Once the agent is in the vicinity of its current box's target, its destination is updated to the next box's target. This continues until it reaches the final target.

Navigation Behaviors

Wandering: occasional random actions deviating from the most direct path

Teleporting: relocates to random location within the region ahead of the agent's current position with a random shift in trajectory towards the target's general direction

Perfect: the "correct", most direct path

With each action the simulation captures an image of the agent's current perspective and stores the next correct action at that location/direction. Using these navigators and the simulated environment, we were able to collect a total of nine datasets with 100,000 image-action pairs each.

MODEL TRAINING

Using fast.ai and the timm library, we trained each of our datasets on six different architectures:

- EfficientNet
- ResNet18
- ViT
- ConvNextV2Base
- ConvNextV2Atto
- MobileNetV4

Our training script uses supervised learning to create a model that, given an image, outputs a prediction from `tensor(left, right, forward)`, informing the agent what action to take.

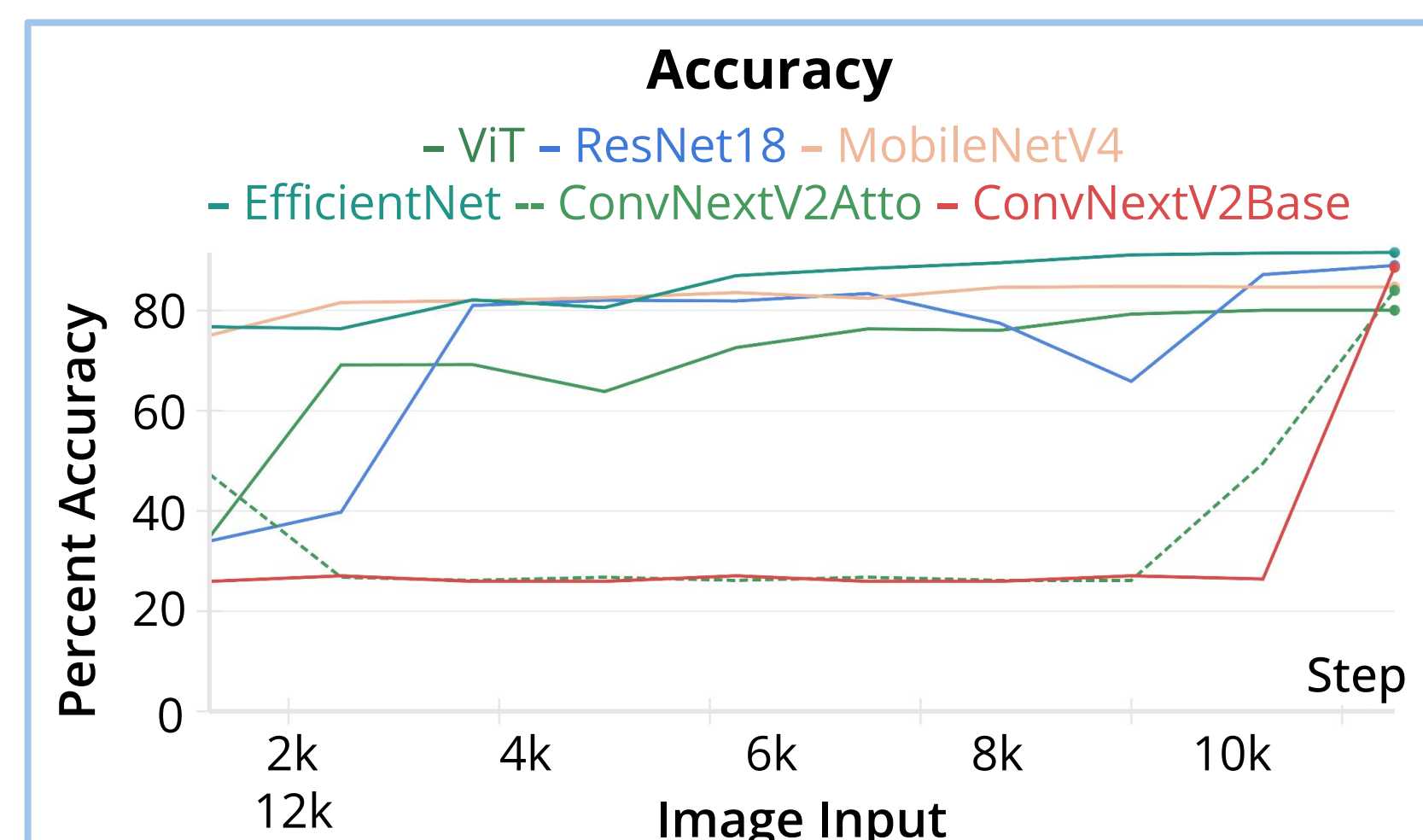


Figure 2: WandB figure displaying the percent accuracy progression during Wandering Static models' training

Once training is complete, the model, and training data are uploaded into Weights & Biases (WandB). WandB then uses this data to create training analysis figures such as the accuracy plot in Figure 2.

RESULTS

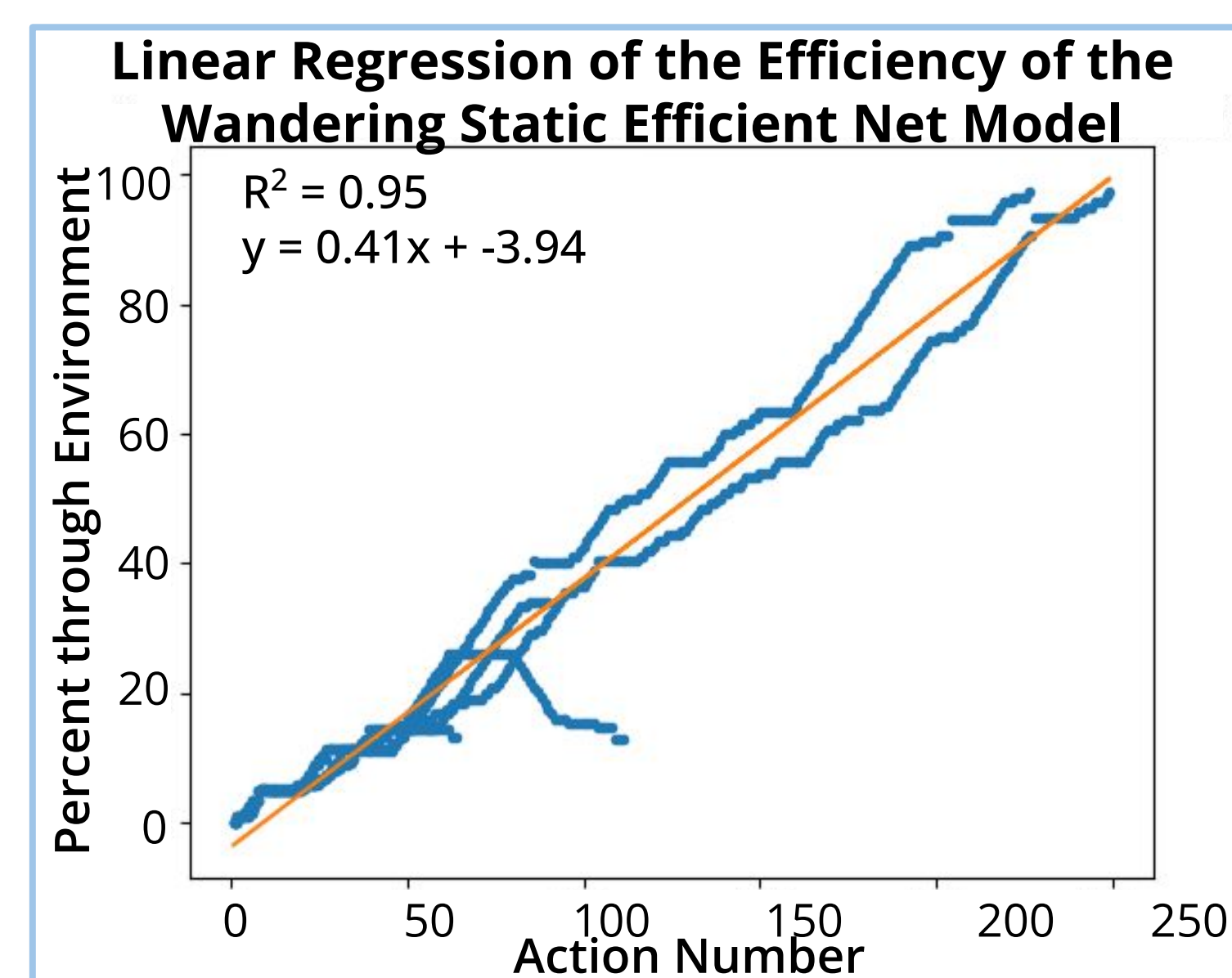


Figure 3: Linear Regression exhibiting the number of actions to achieve progress in Wandering Static Efficient Net model

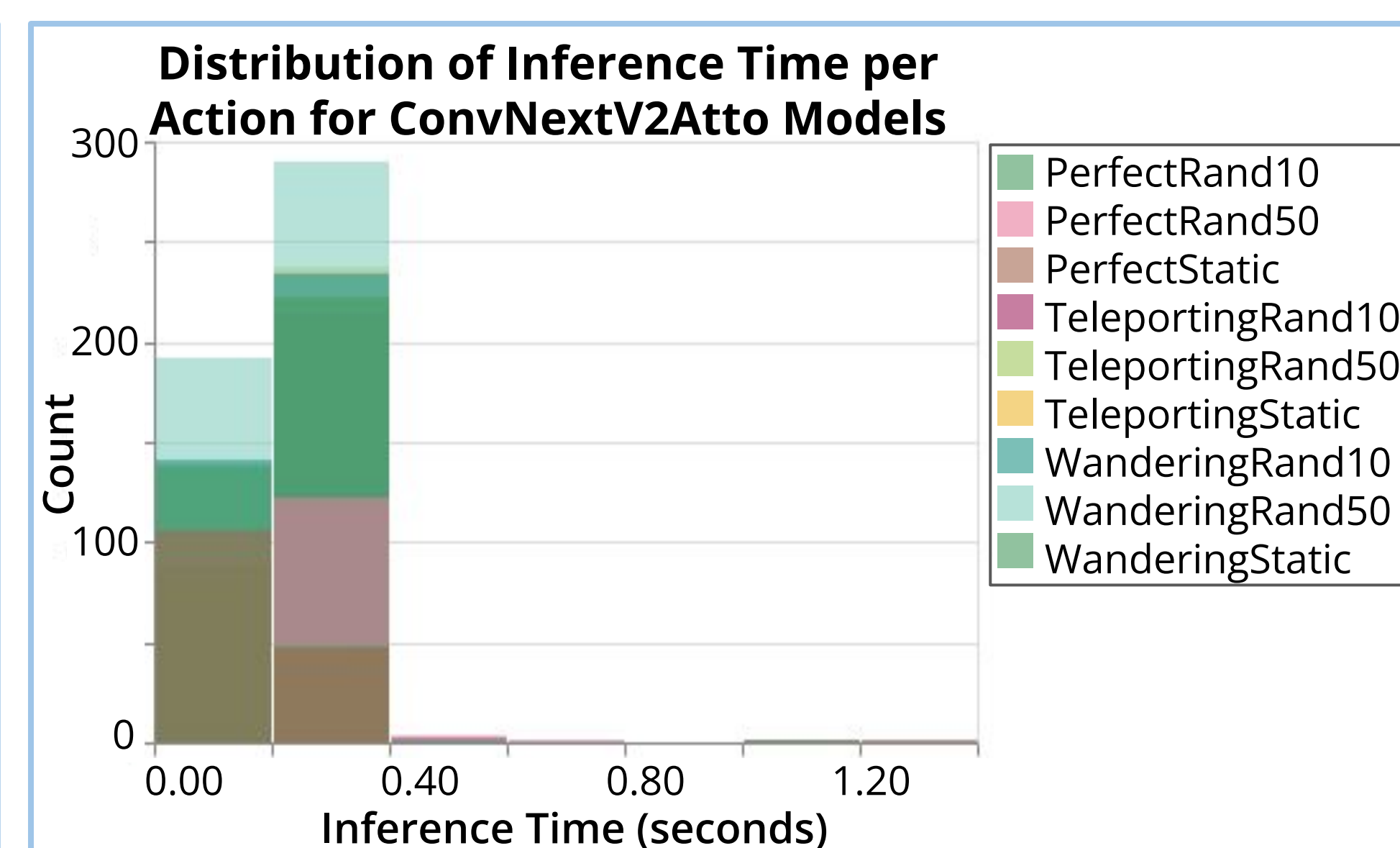


Figure 4: Distribution of the models trained on ConvNextV2Atto's decision-making speed per action

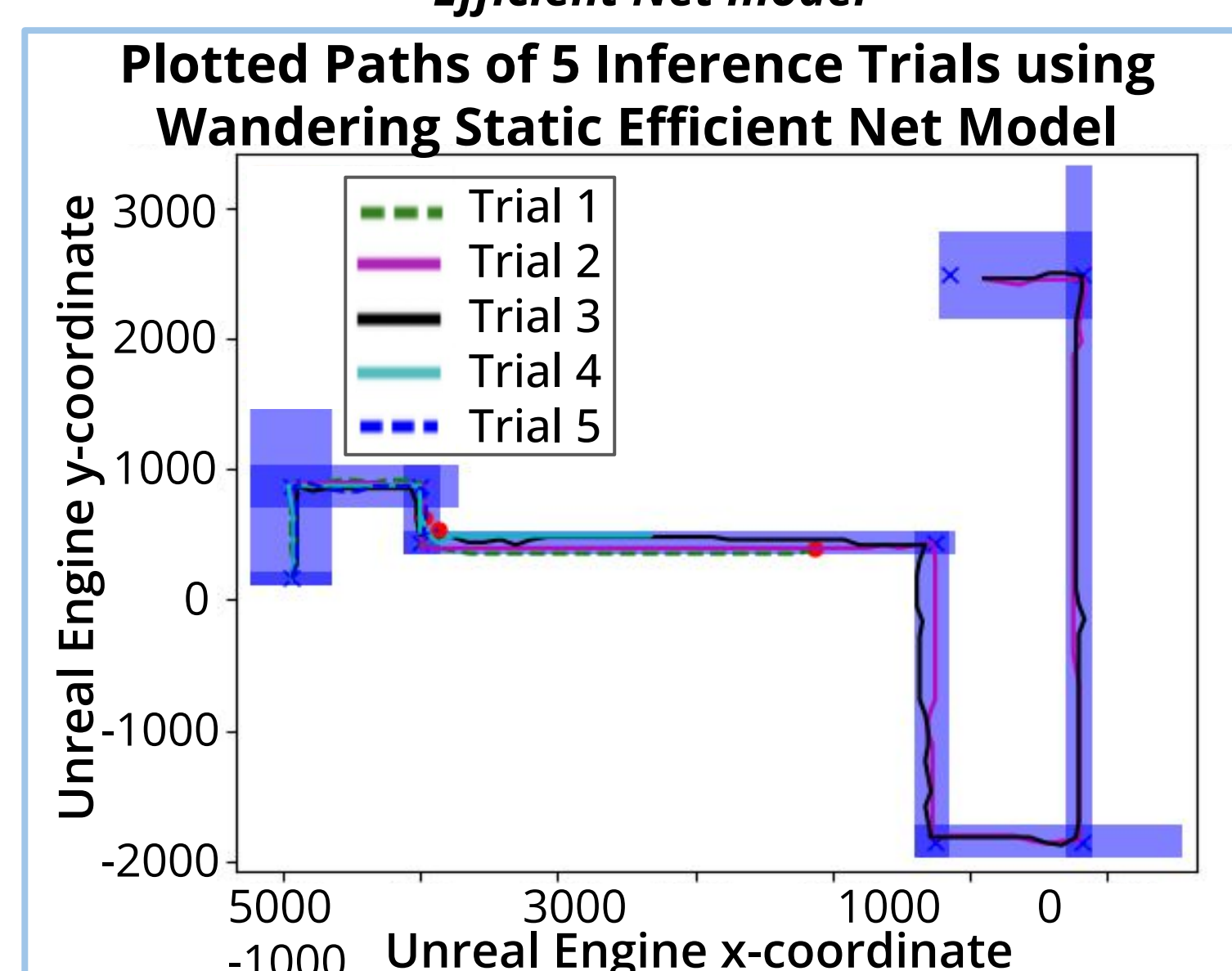


Figure 5: Path of each trial in a run to analyze consistency of Wandering Static Efficient Net model

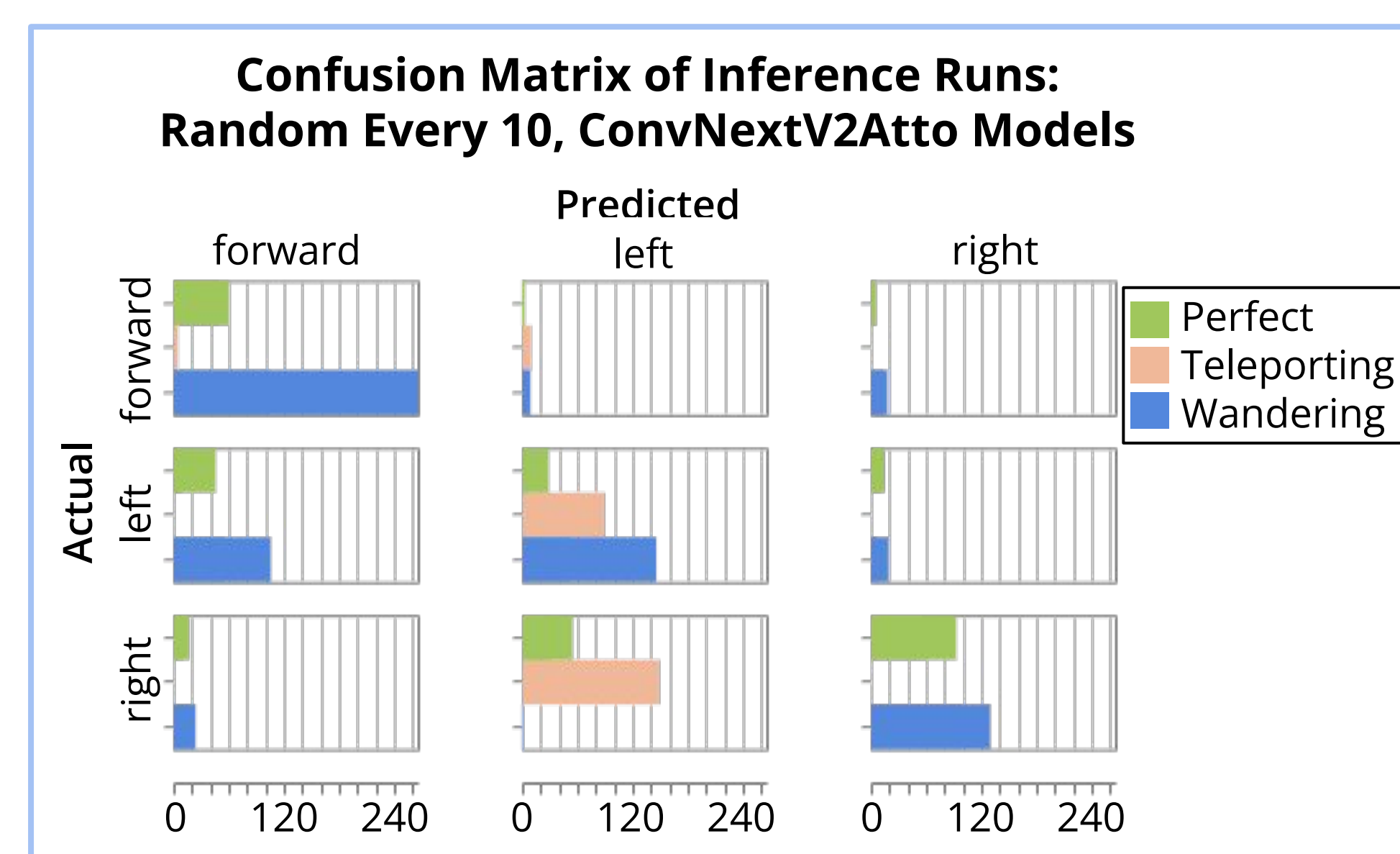


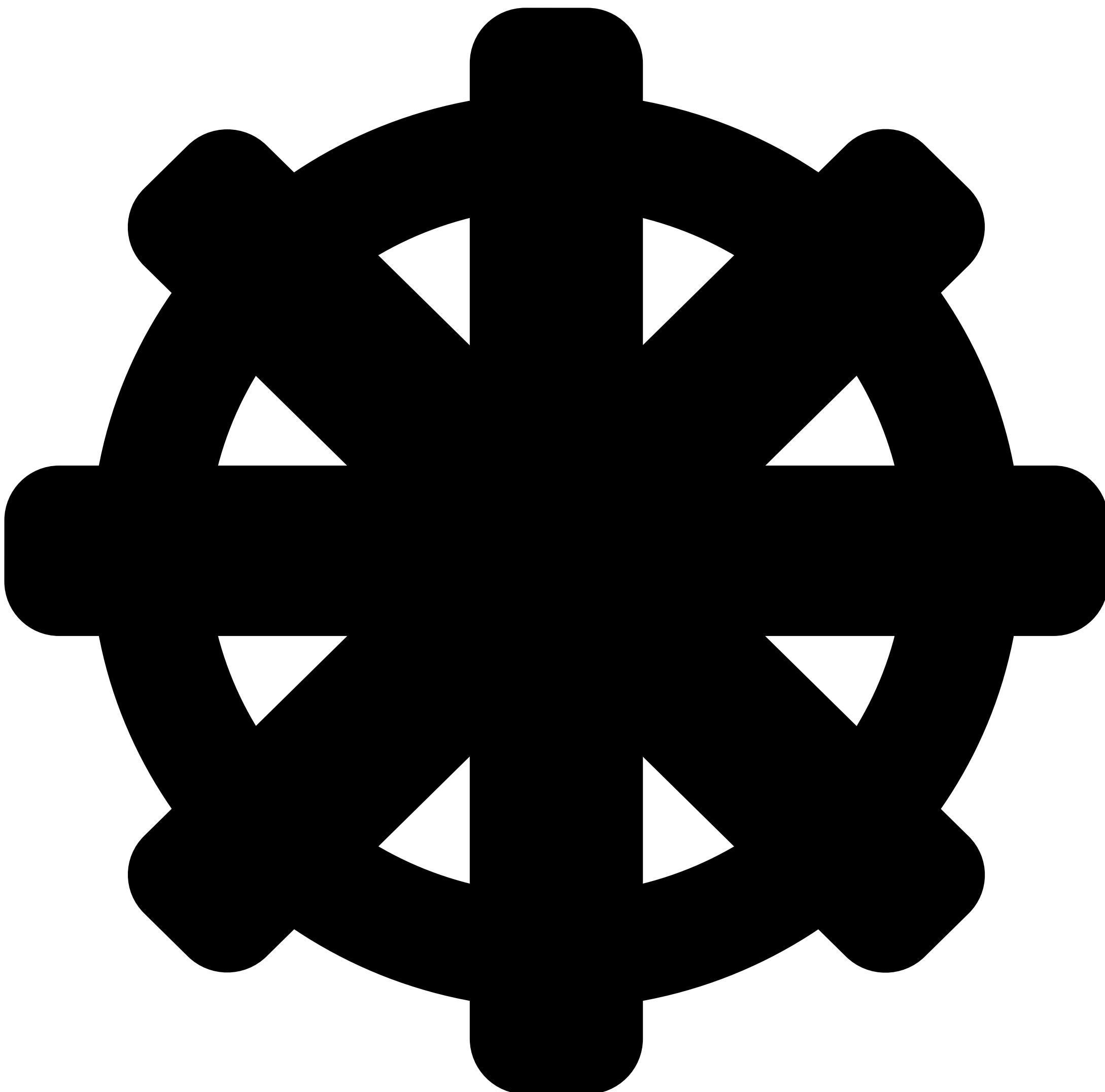
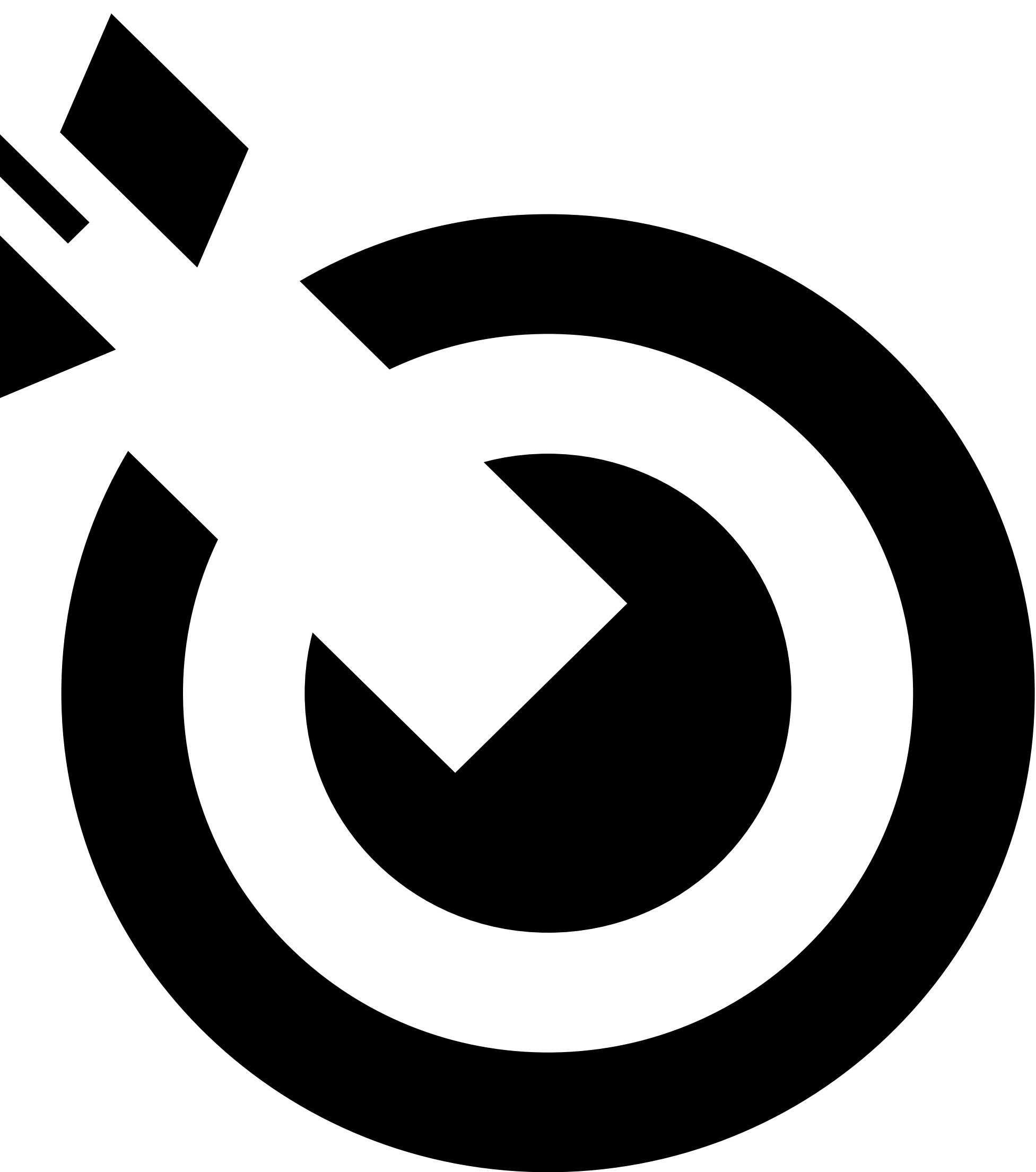
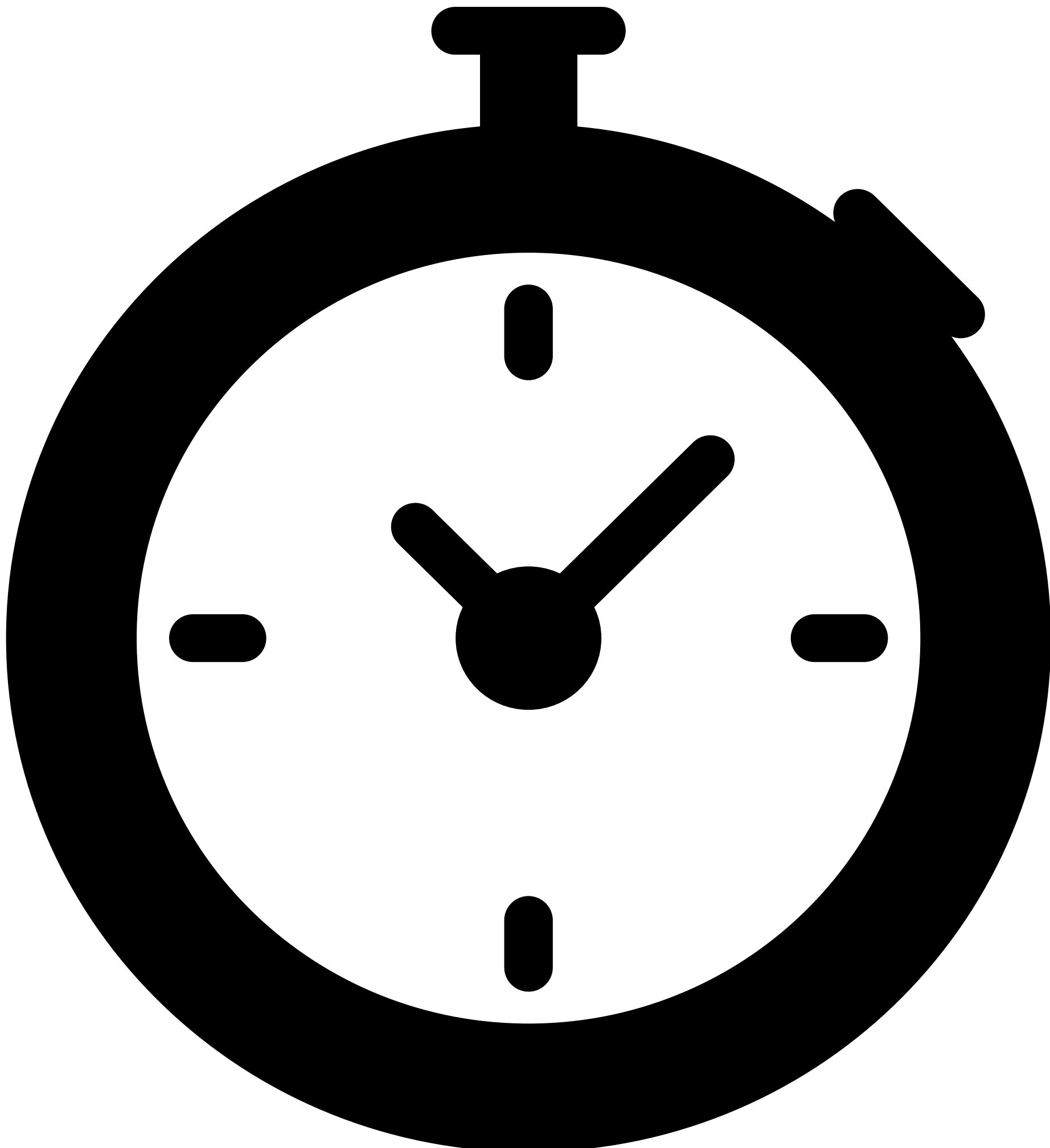
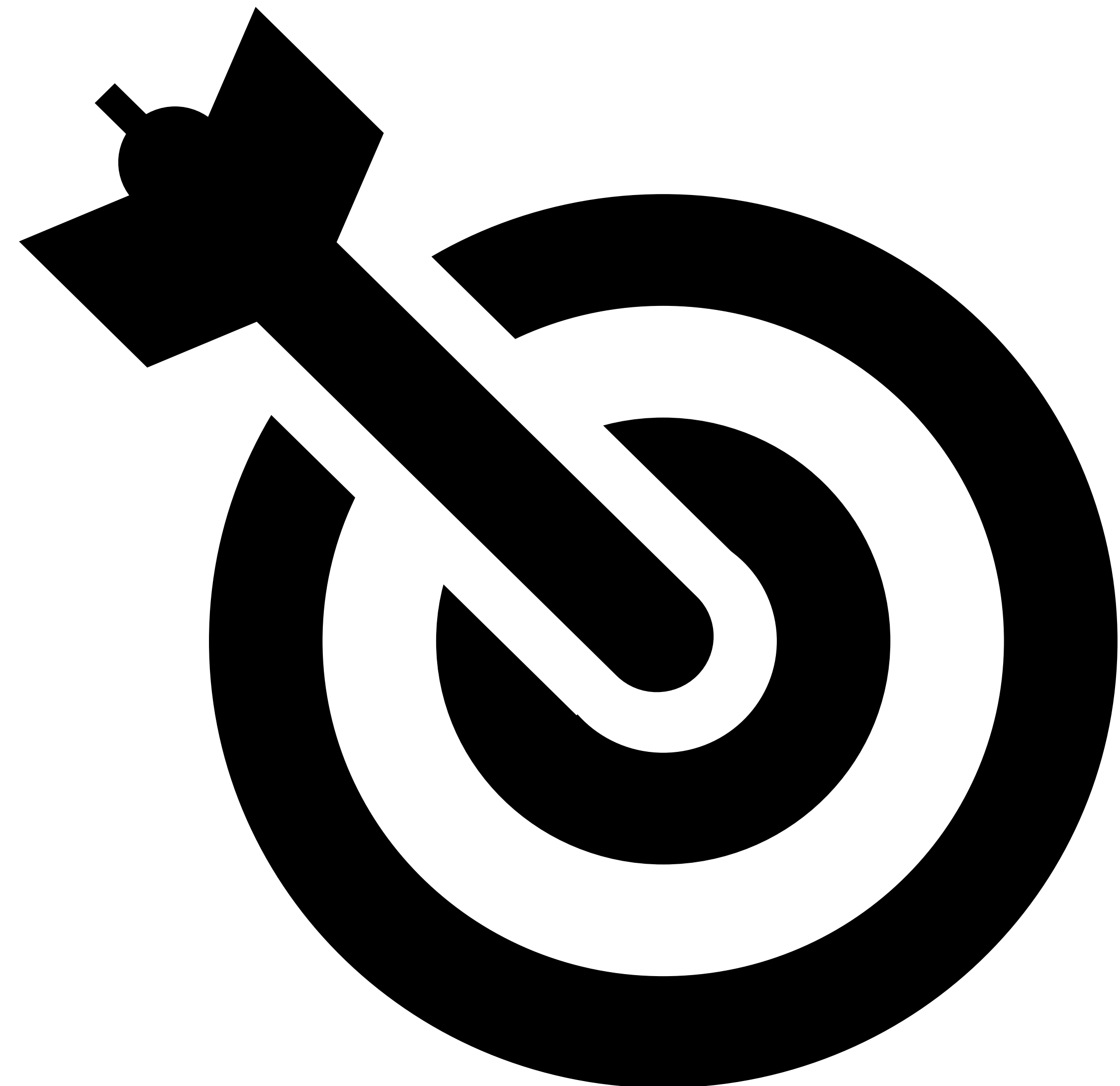
Figure 6: Confusion Matrix telling which actions the ConvNextV2Atto (trained on randomized textures) models is best at predicting

NEXT STEPS

Diverse Datasets: We hope to diversify our dataset collection by slightly altering the locations of the targets after each trial, so the navigators to traverse different paths.

Data Augmentation: In pursuit of deploying these models into our prototype robot, we plan to use data augmentation during training to improve the models' performance in a real world setting.

Step Out of the Simulation: Along the way, we will evaluate the models in real life using our prototype robot in the environment that our simulation was based off of.



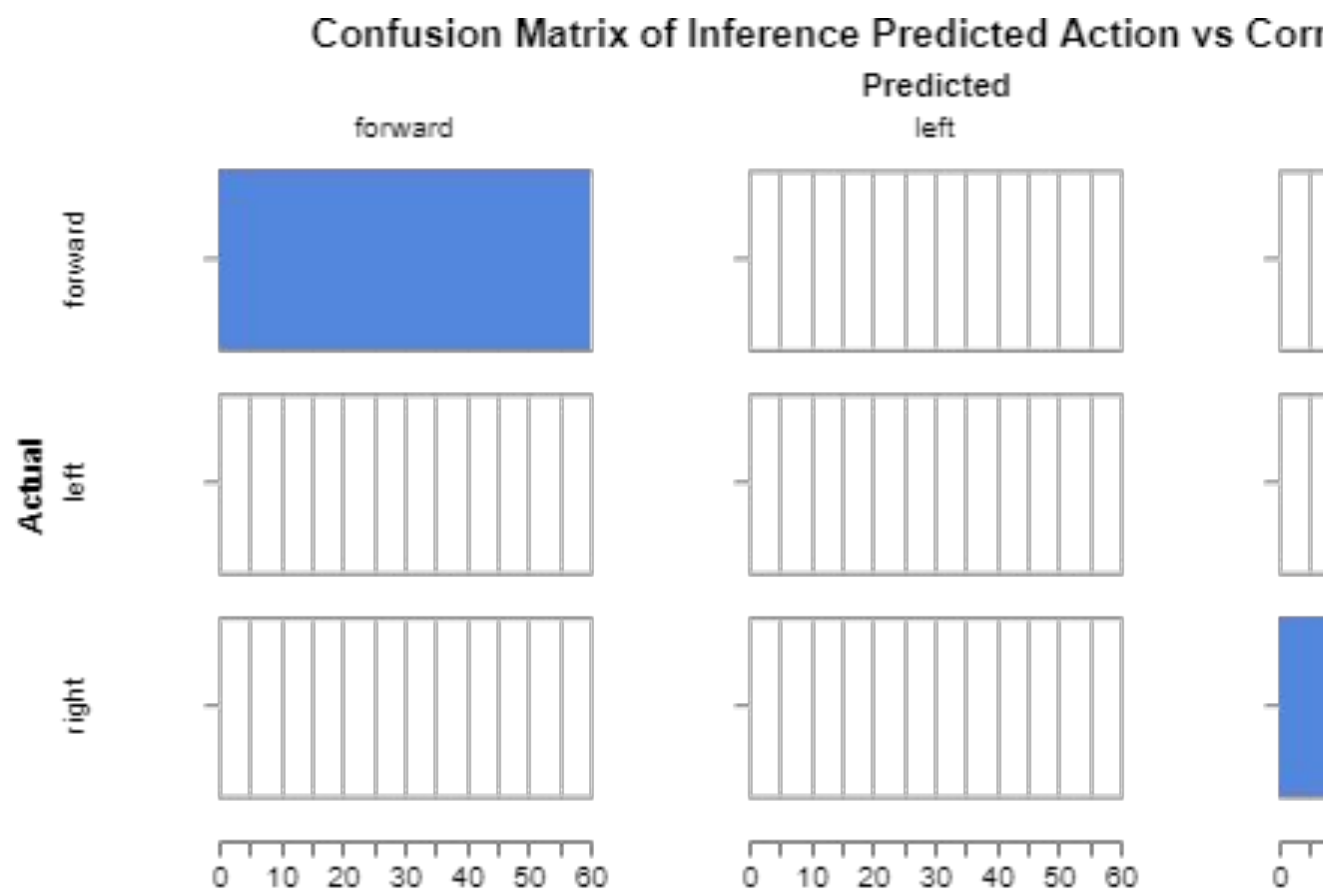
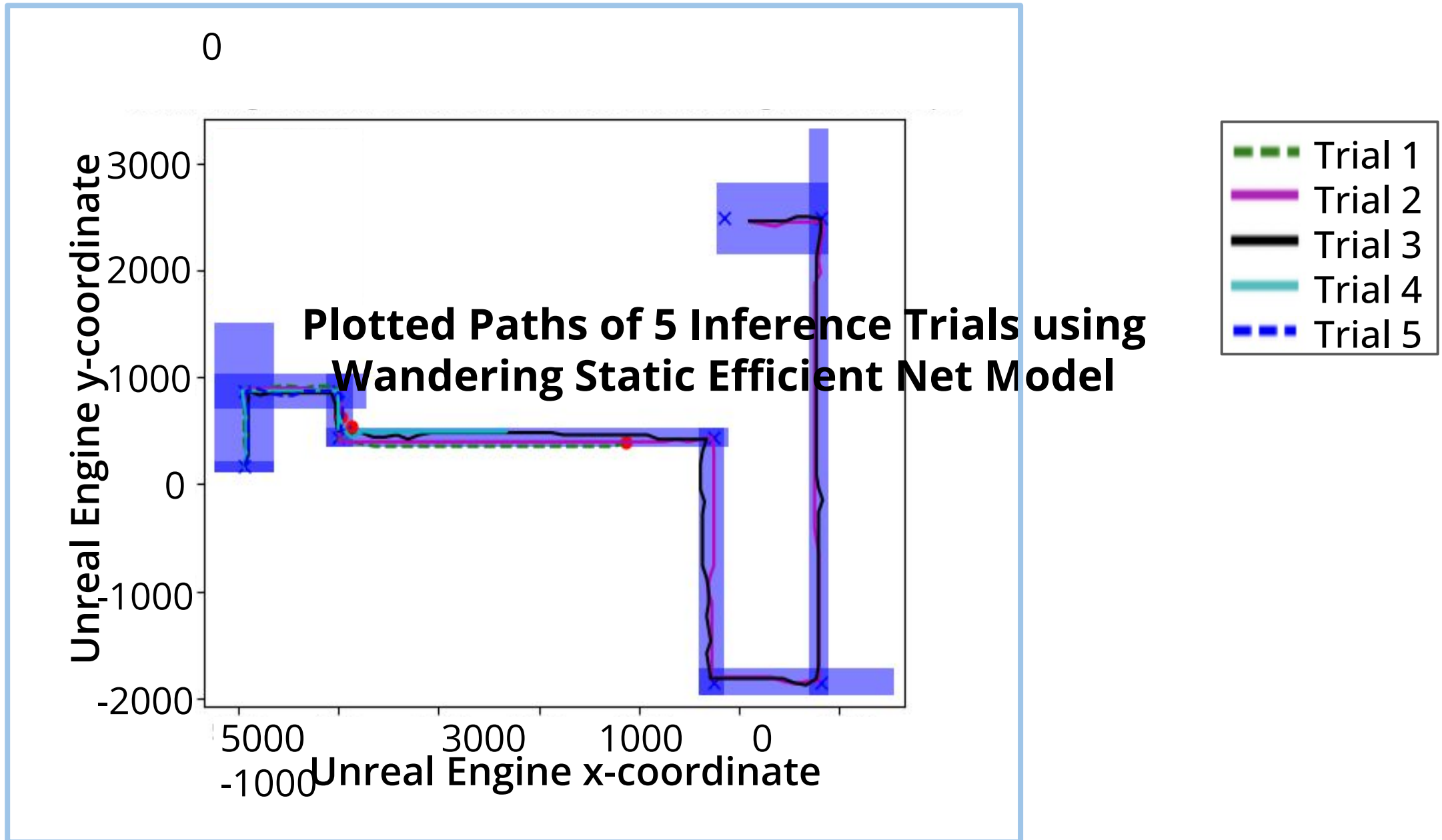
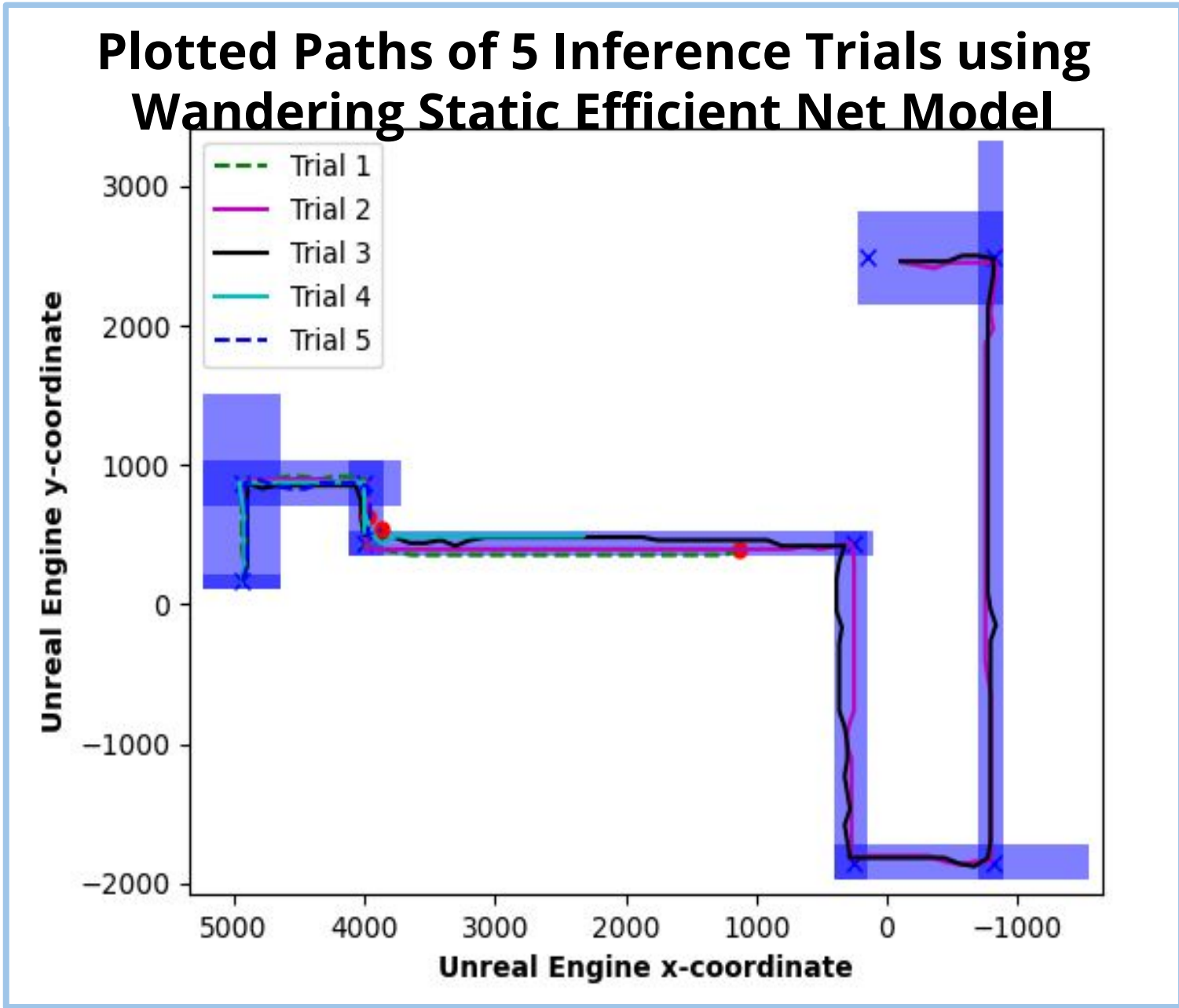
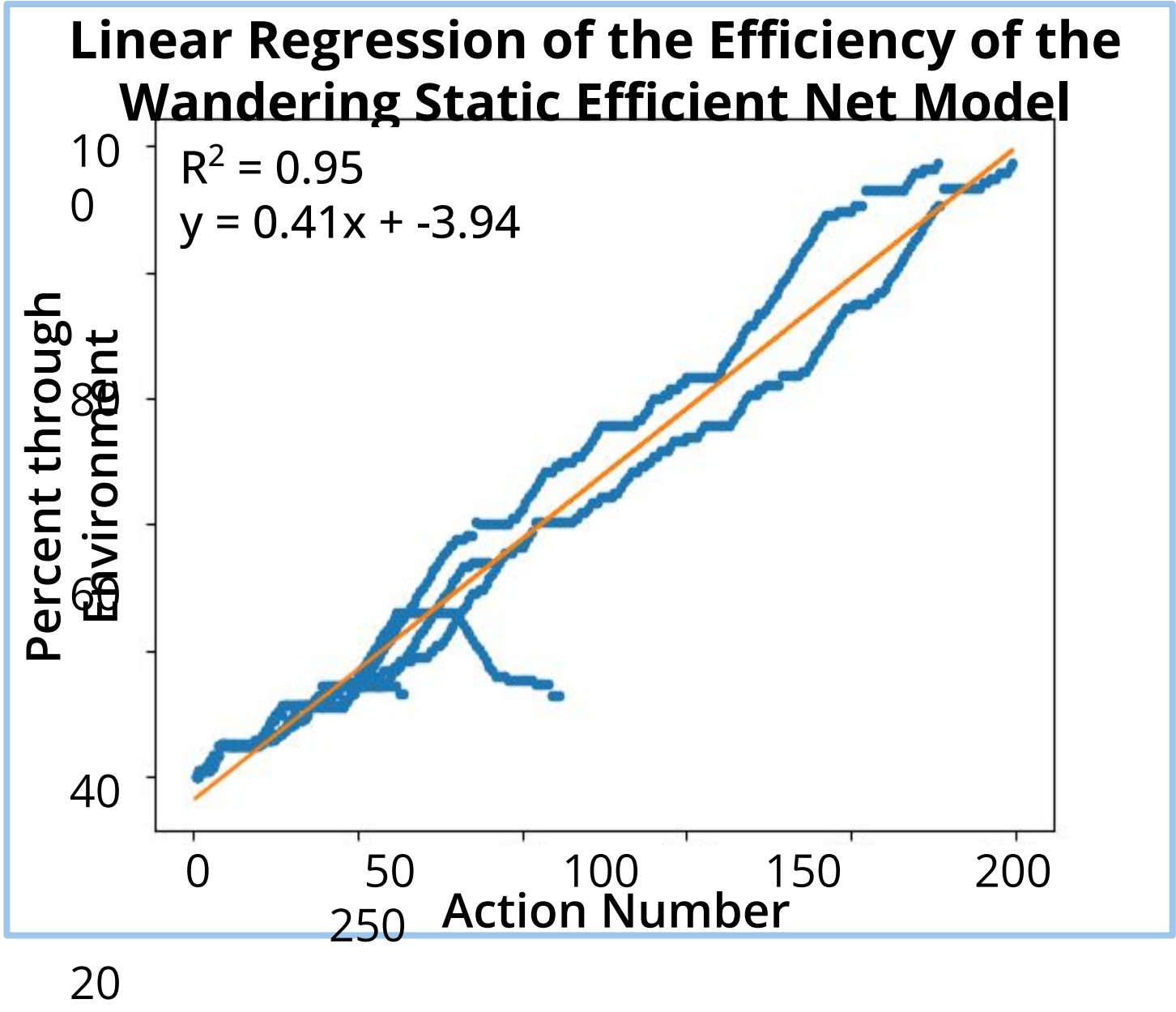
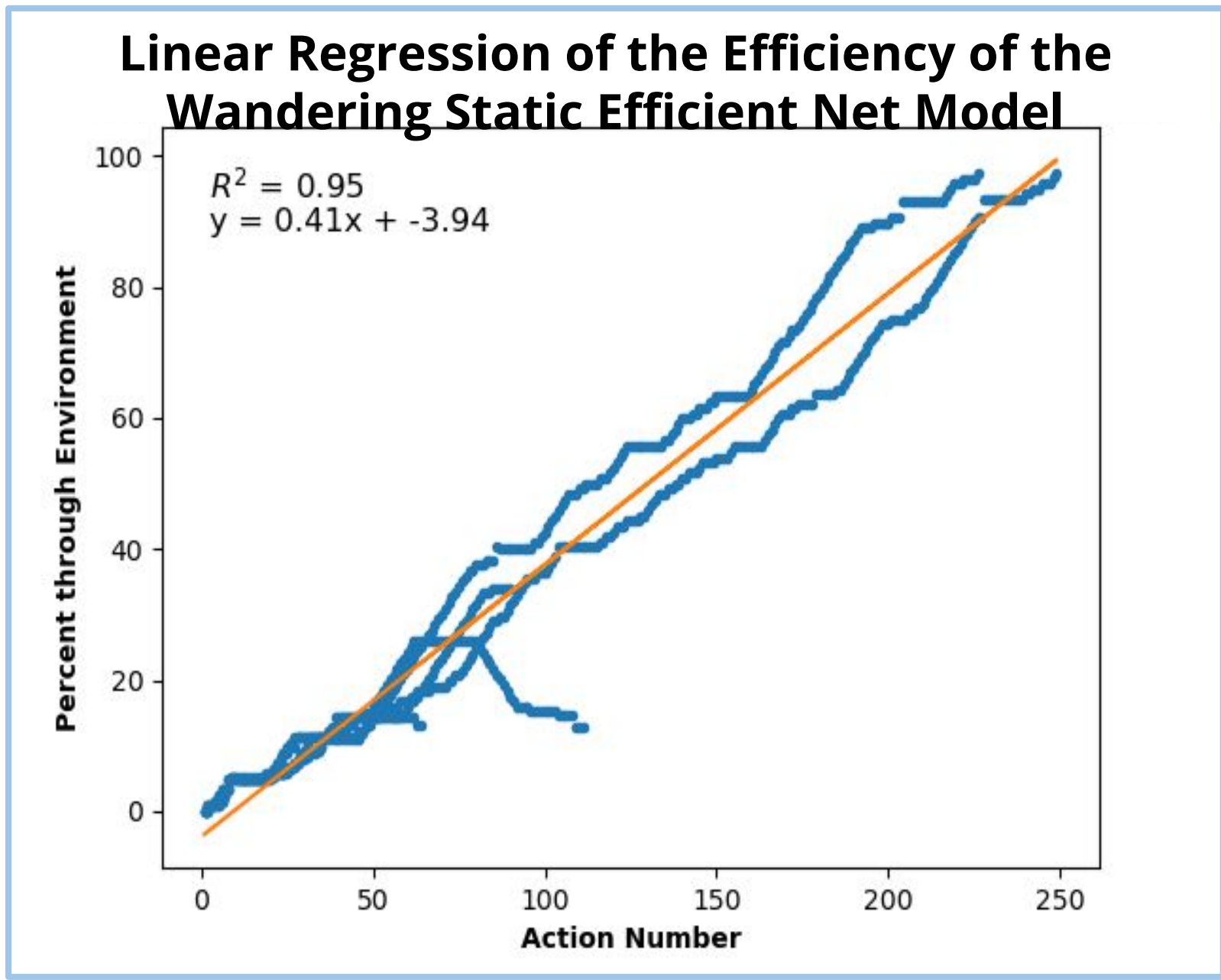
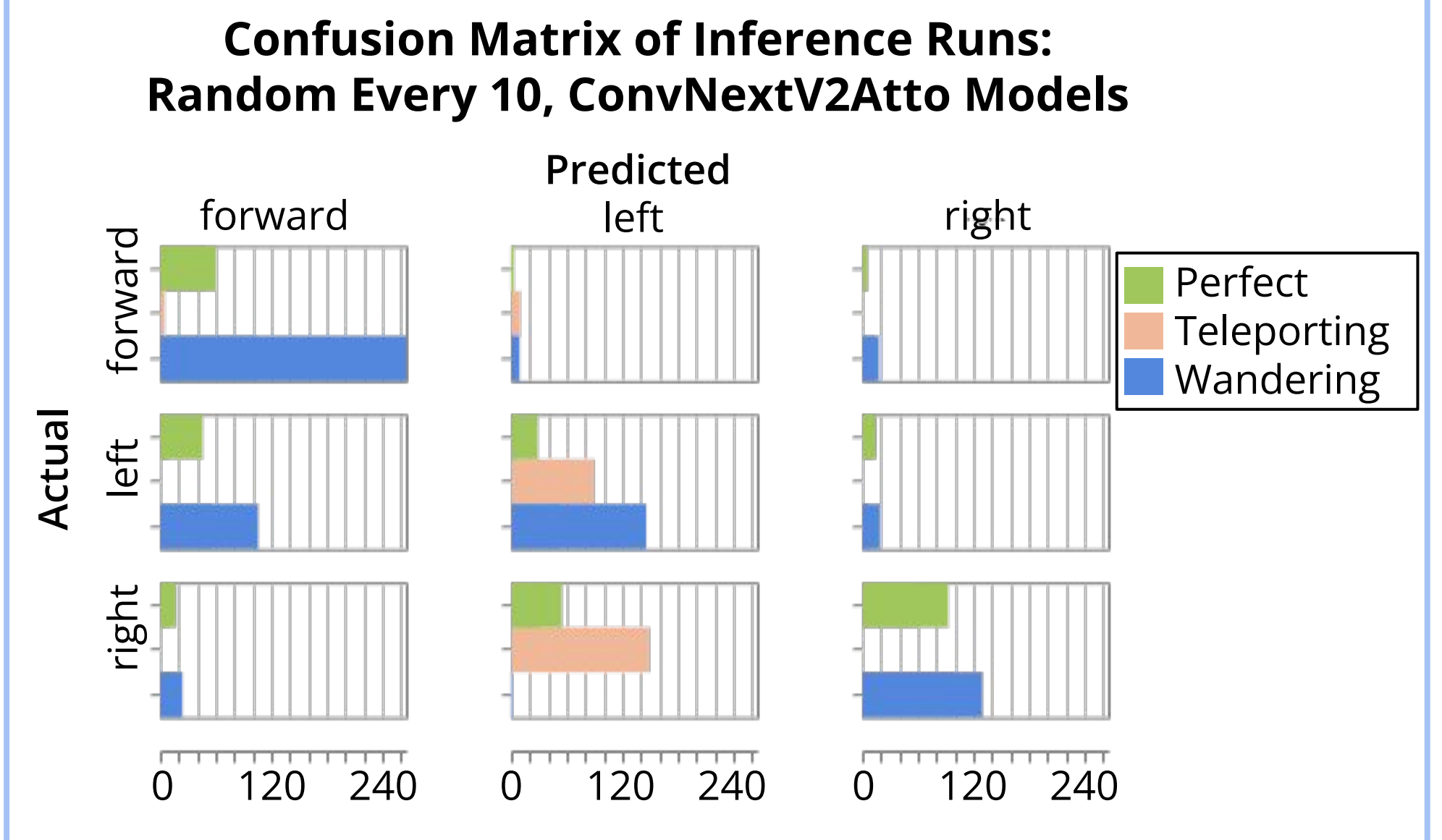
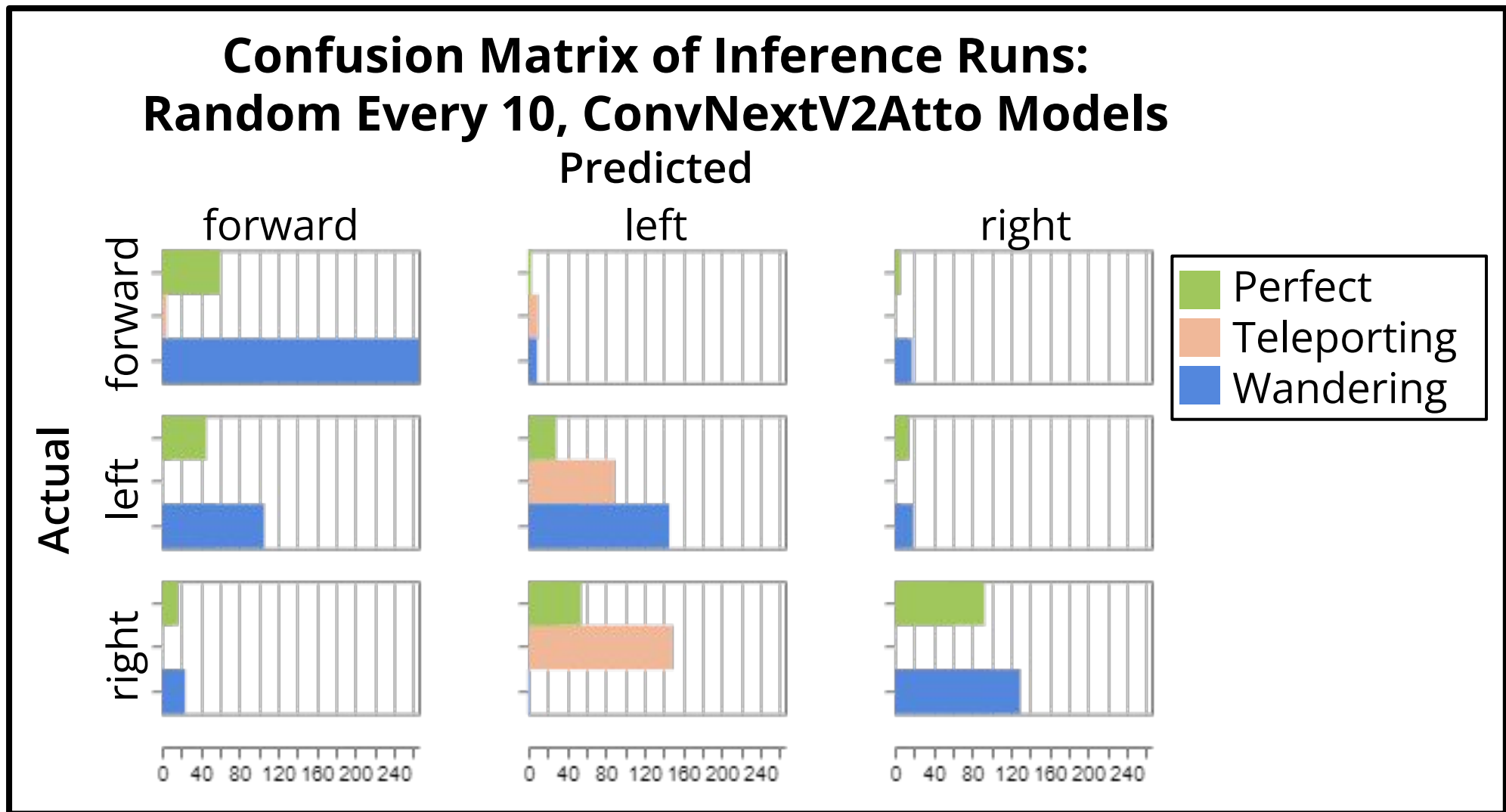
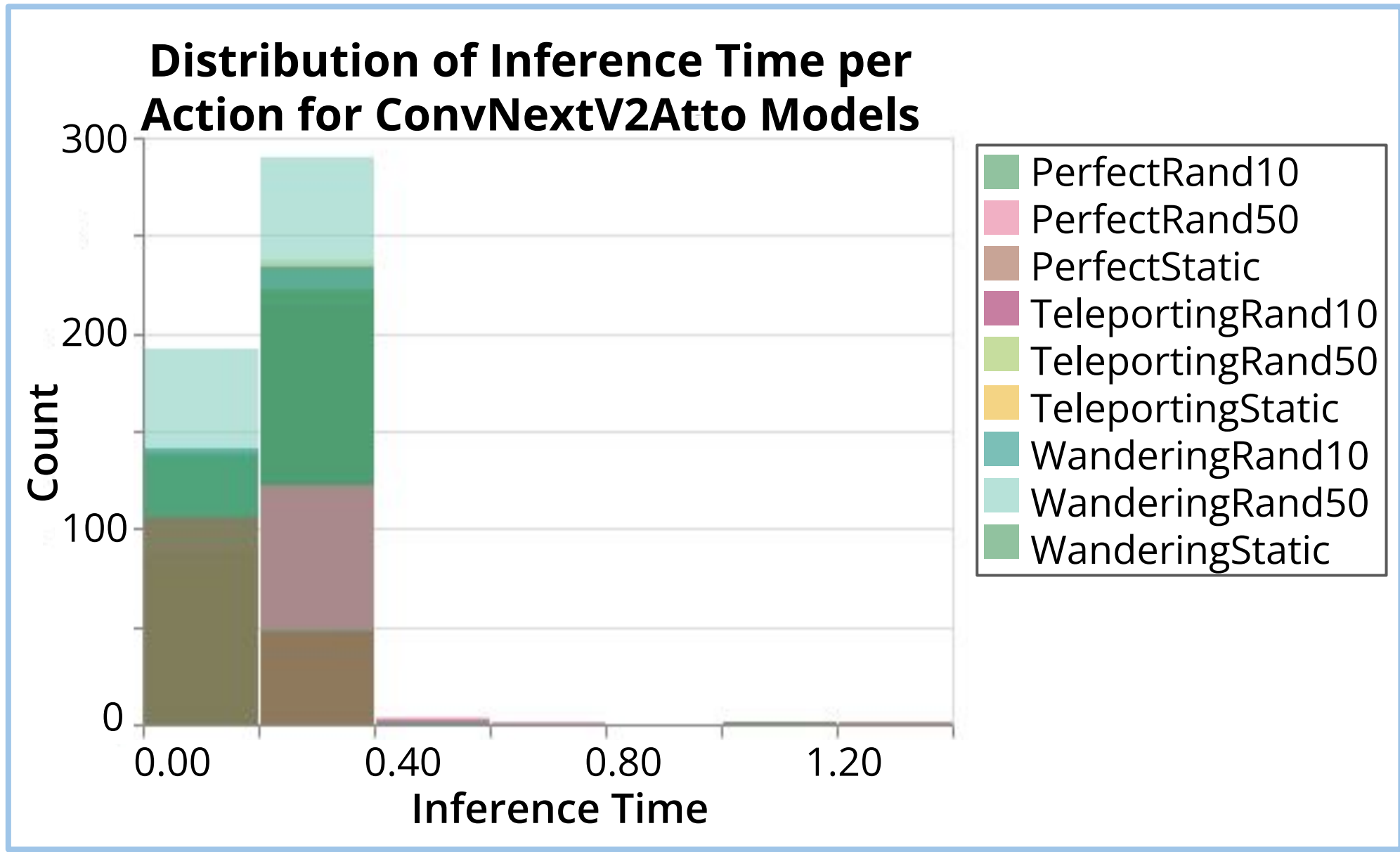
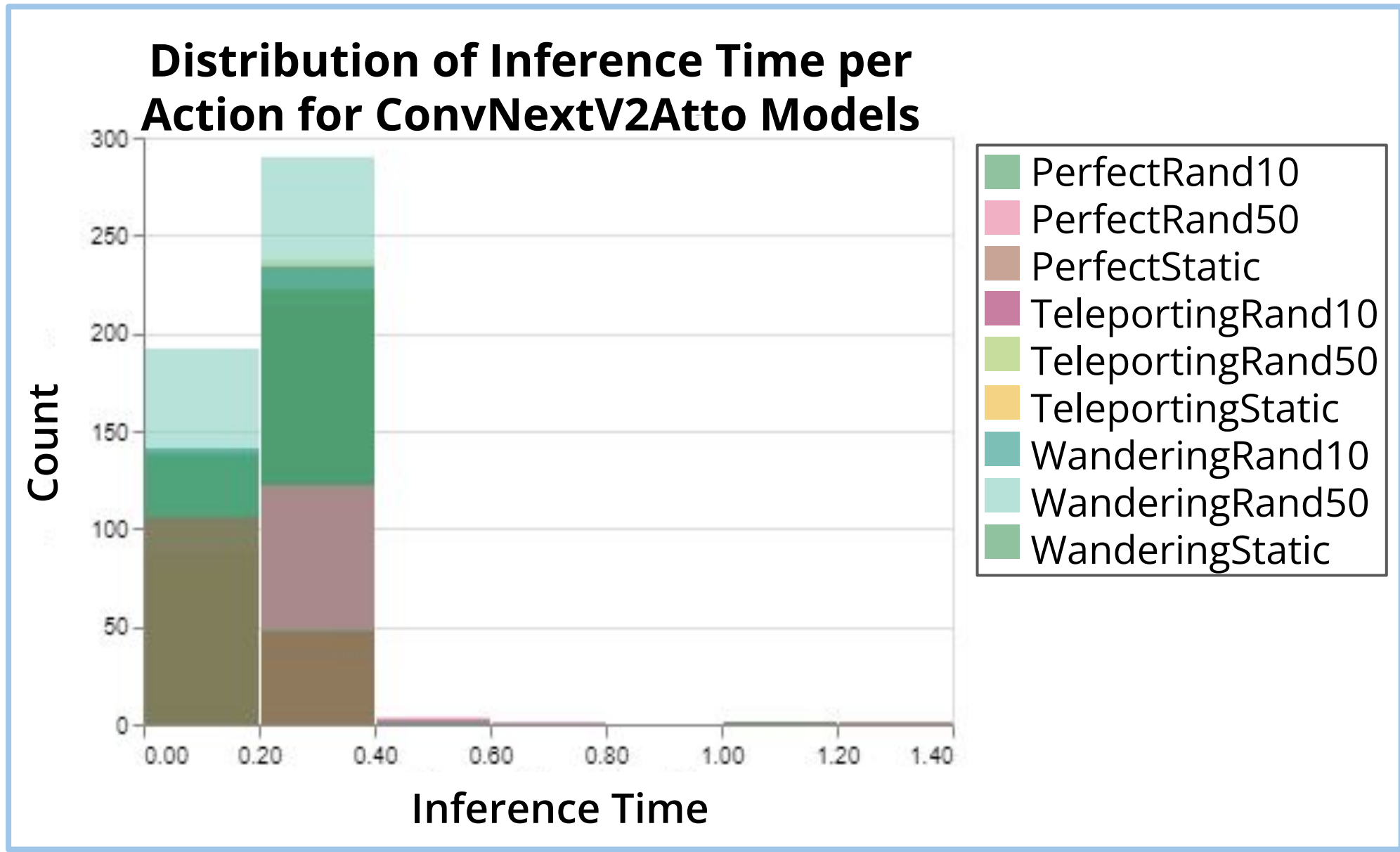


Figure __: Confusion Matrix telling which action model is best at predicting

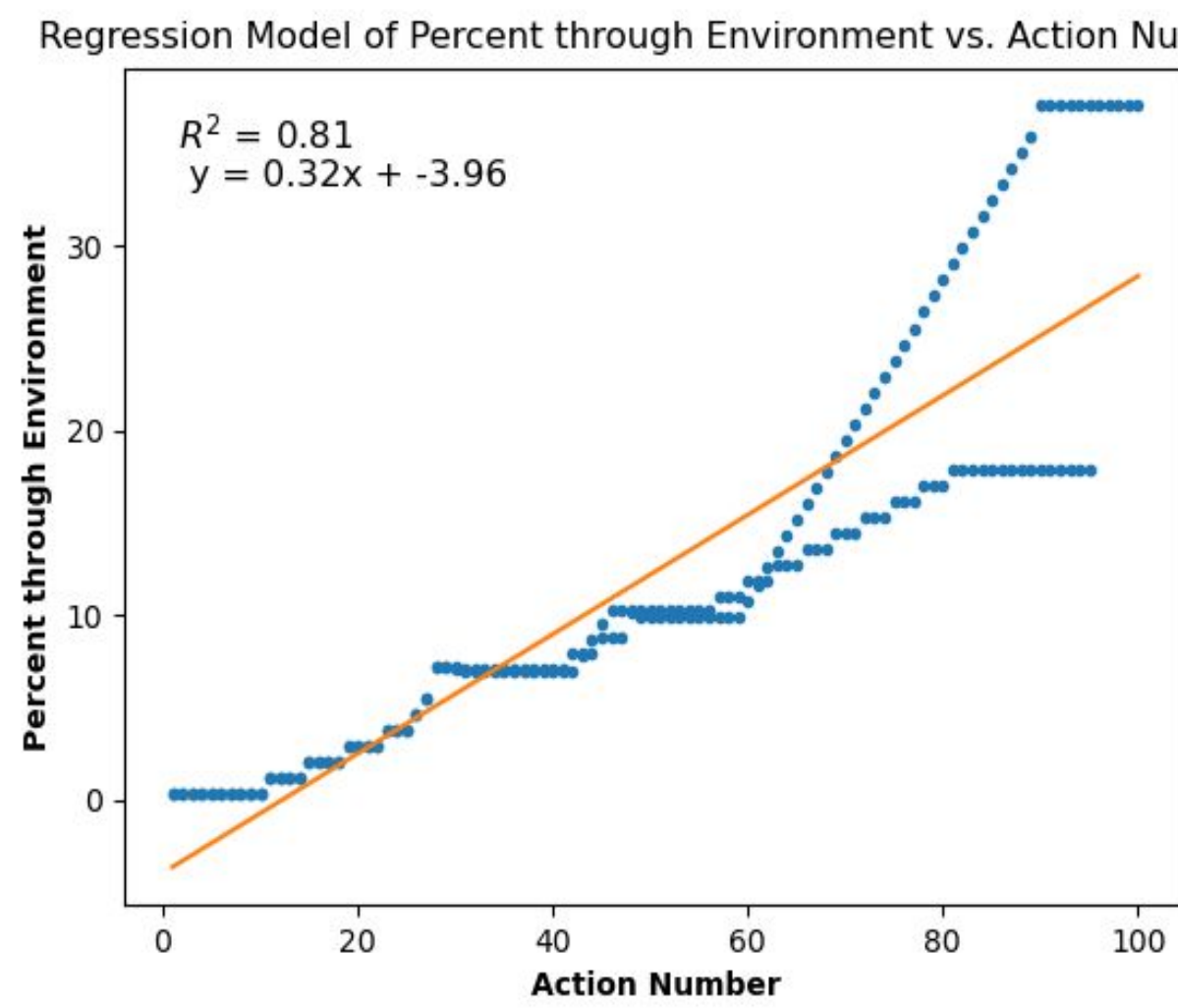
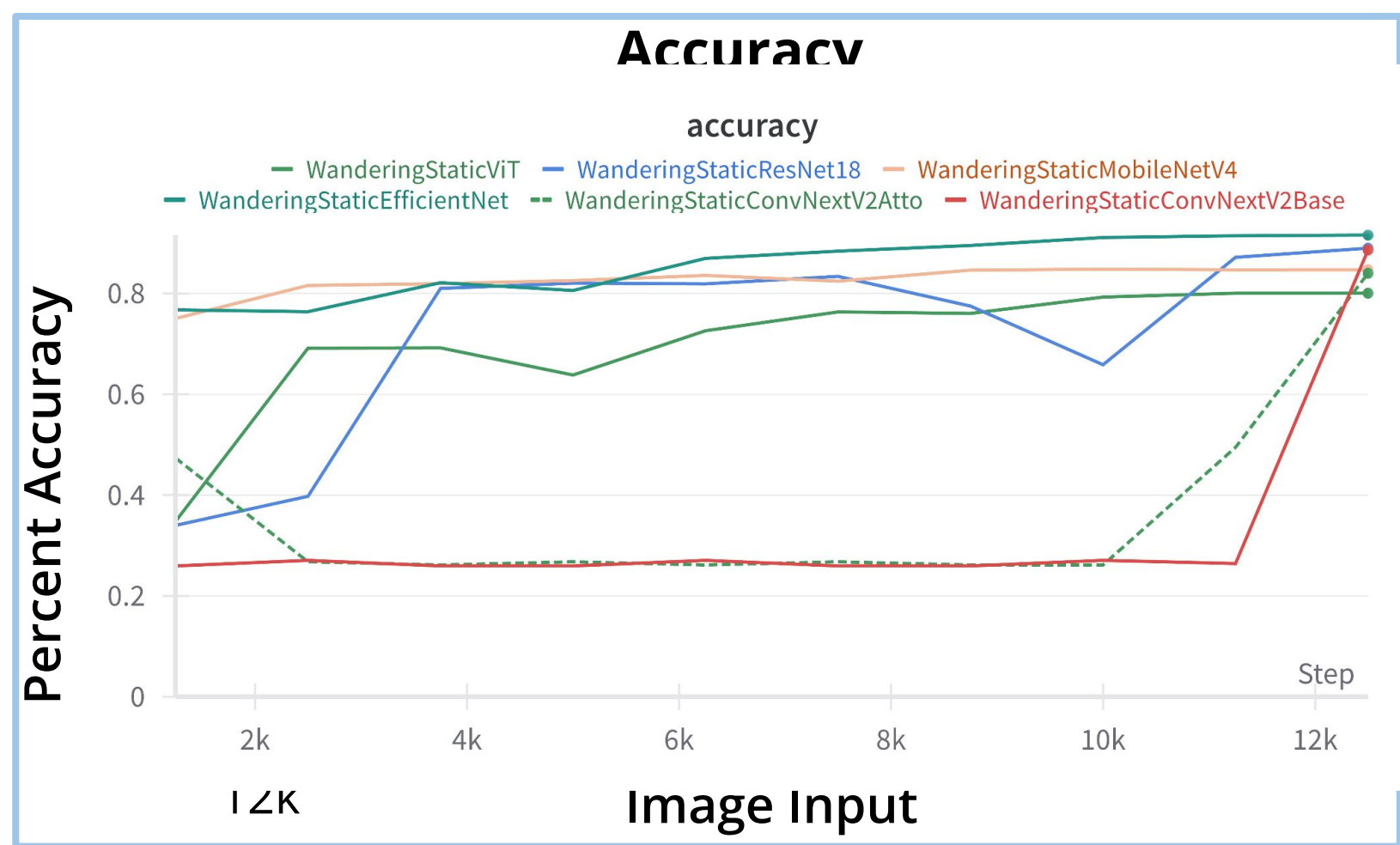


Figure __: Linear regression exhibiting how many the __ model takes to complete the enviro